

DPT Exercises

Exercise sheet 9

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Exercise 1 (sum of waiting times). The time (in minutes) a customer i has to wait for their coffee at a specialty coffee shop is modeled by an exponential random variable $X_i \sim \text{Exp}(\lambda)$ with $\lambda = 0.5$ (so the average wait is $1/\lambda = 2$ minutes). Let $S_{36} = \sum_{i=1}^{36} X_i$ be the total time required to serve 36 independent customers.

- (i) Find the characteristic function of a single waiting time X_i .
- (ii) Using the result from (i), what is the exact distribution of the total waiting time S_{36} ? State its name and parameters.
[Hint: Look at Example 2.50 of the lecture.]
- (iii) Use the clt to approximate the probability that the total waiting time for 36 customers exceeds 80 minutes.
[Hint: You will need the mean and variance of an $\text{Exp}(0.5)$ distribution. For the clt, you may use that for a standard normal RVRV Z , we have $P(Z > 2/3) \approx 0.2525$.]

Solution:

- (i) For $X_i \sim \text{Exp}(\lambda)$, the characteristic function is $\phi_{X_i}(t) = \mathbb{E}[e^{itX_i}] = \int_0^\infty e^{itx} \lambda e^{-\lambda x} dx$. Which we have already seen to be

$$\phi_{X_i}(t) = \lambda \int_0^\infty e^{-(\lambda - it)x} dx = \lambda \left[\frac{e^{-(\lambda - it)x}}{-(\lambda - it)} \right]_0^\infty = \lambda \left(0 - \frac{1}{-(\lambda - it)} \right) = \frac{\lambda}{\lambda - it} = \frac{1}{1 - it/\lambda}.$$

With $\lambda = 0.5$, this is $\phi_{X_i}(t) = (1 - 2it)^{-1}$.

- (ii) The characteristic function of the sum $S_{36} = \sum_{i=1}^{36} X_i$ of iid variables is the product of their individual characteristic functions

$$\phi_{S_{36}}(t) = \prod_{i=1}^{36} \phi_{X_i}(t) = (\phi_{X_i}(t))^{36} = \left(\frac{1}{1 - it/0.5} \right)^{36}.$$

This is the characteristic function of a Gamma distribution $\Gamma(\alpha, \beta)$ which is $(1 - it/\beta)^{-\alpha}$. By comparing the forms, we see that $\alpha = 36$ and $\beta = \lambda = 0.5$. Thus, $S_{36} \sim \Gamma(36, 0.5)$.

- (iii) For $X_i \sim \text{Exp}(0.5)$ we have $\mathbb{E}[X_i] = 1/\lambda = 2$ and $\text{var}[X_i] = 1/\lambda^2 = 4$. For the sum S_{36} , we then find $\mathbb{E}[S_{36}] = n \cdot \mathbb{E}[X_i] = 72$ and $\text{var}[S_{36}] = n \cdot \text{var}[X_i] = 144$. The standard deviation is $\sigma_{S_{36}} = \sqrt{144} = 12$ and by the clt approximately $S_{36} \sim \mathcal{N}(72, 144)$. To approximate $P(S_{36} > 80)$ we standardize via

$$z = \frac{80 - \mathbb{E}[S_{36}]}{\sqrt{\text{var}[S_{36}]}} = \frac{80 - 72}{12} = \frac{2}{3}.$$

So, $P(S_{36} > 80) \approx P(Z > 2/3)$, where $Z \sim \mathcal{N}(0, 1)$. $P(Z > 2/3) = 1 - \Phi(2/3) \approx 1 - 0.7475 = 0.2525$. There is approximately a 25.25% chance the total wait time exceeds 80 minutes.

Exercise 2 (quality control in manufacturing). A factory produces resistors. The resistance of each resistor is intended to be 100 ohms. Due to manufacturing variability, the actual resistance values are iid random variables with mean $\mu = 100$ ohms and standard deviation $\sigma = 2$ ohms. A batch consists of $n = 64$ resistors. A batch is rejected if the average resistance of the 64 resistors is outside the range $[99.5, 100.5]$ ohms.

- (i) Let $\bar{X}_{64}$ be the average resistance of a batch of 64 resistors. What are the mean and variance of $\bar{X}_{64}$?

- (ii) Use Chebyshev's inequality to find an upper bound on the probability that a batch is rejected.
- (iii) Use the clt to approximate the probability that a batch is rejected.
[Hint: You may use that for a standard normal RVRV Z , we have $P(Z > 2) \approx 0.0228$.]

Solution:

- (i) Let X_i be the resistance of the i -th resistor, with $\mathbb{E}[X_i] = 100$ and $\text{var}[X_i] = \sigma^2 = 2^2 = 4$. For the sample mean $\bar{X}_{64} = \frac{1}{64} \sum_{i=1}^{64} X_i$ we have $\mathbb{E}[\bar{X}_{64}] = \mu = 100$ and $\text{var}[\bar{X}_{64}] = \frac{\sigma^2}{n} = \frac{1}{16}$. The standard deviation of the sample mean is $\sigma_{\bar{X}_{64}} = \sqrt{1/16} = 1/4$.
- (ii) A batch being rejected is the event $|\bar{X}_{64} - 100| > 0.5$. We use Chebyshev's inequality with $\epsilon = 0.5$ and the properties of $\bar{X}_{64}$ to find

$$P(|\bar{X}_{64} - 100| > 0.5) \leq \frac{\text{var}[\bar{X}_{64}]}{\epsilon^2} = \frac{1/16}{(0.5)^2} = \frac{1}{4}.$$

The probability of rejecting a batch is at most 25%.

- (iii) By the clt, $\bar{X}_{64} \approx \mathcal{N}(\mu, \sigma^2/n) = \mathcal{N}(100, 1/16)$. The rejection probability is $P(\bar{X}_{64} < 99.5) + P(\bar{X}_{64} > 100.5)$. With standardization $Z = \frac{\bar{X}_{64} - 100}{1/4}$, the lower tail is $z_1 = \frac{99.5 - 100}{0.25} = -2$ and the upper tail is $z_2 = \frac{100.5 - 100}{0.25} = 2$. The probability of rejection is thus approximately

$$P(Z < -2) + P(Z > 2) = 2 \cdot P(Z > 2) = 2 \cdot 0.0228 = 0.0456,$$

using the hint (or any programming language/wolframalpha/etc). The clt approximation suggests that about 4.56% of batches will be rejected—much fewer than the 25% upper bound from Chebyshev's inequality.

Exercise 3 (transformation and convergence in distribution). Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of RVRVs with $X_n \sim \text{Unif}(0, 1/n)$ for each $n \in \mathbb{N}_{>0}$. Let $Y_n = nX_n$.

- (i) For a fixed n , find the pdf $p_{Y_n}(y)$ of Y_n .
[Hint: Use the change of variables formula.]
- (ii) Find the cdf $F_{Y_n}(y)$ of Y_n .
- (iii) Determine the limiting cdf $F_Y(y) = \lim_{n \rightarrow \infty} F_{Y_n}(y)$.
- (iv) To what RVRV Y does Y_n converge in distribution? Specify its distribution.

Solution:

- (i) The pdf of $X_n \sim \text{Unif}(0, 1/n)$ is $p_{X_n}(x) = n$ for $x \in (0, 1/n)$ and 0 otherwise. You can use the formula for affine transformations here, but let's go through it in detail once. Let $g(x) = nx$ so $Y_n = g(X_n)$. The inverse transformation is $g^{-1}(y) = y/n$. The derivative is $\frac{d}{dy}g^{-1}(y) = 1/n$. The support of X_n is $S_{X_n} = (0, 1/n)$. The support of Y_n is $g(S_{X_n}) = (0, 1)$. Using the change of variables formula for $y \in (0, 1)$ we find

$$p_{Y_n}(y) = p_{X_n}(g^{-1}(y)) \left| \frac{d}{dy}g^{-1}(y) \right| = p_{X_n}(y/n) \cdot \left| \frac{1}{n} \right|.$$

Since $y \in (0, 1)$, $y/n \in (0, 1/n)$, so $p_{X_n}(y/n) = n$. Thus, $p_{Y_n}(y) = n \cdot \frac{1}{n} = 1$ for $y \in (0, 1)$. So, $p_{Y_n}(y) = 1$ if $y \in (0, 1)$ and 0 otherwise. This is the pdf of a $\text{Unif}(0, 1)$ distribution.

Could you have guessed that intuitively?

- (ii) The cdf of a $\text{Unif}(0, 1)$ is just

$$F_{Y_n}(y) = \begin{cases} 0 & \text{if } y < 0 \\ y & \text{if } 0 \leq y \leq 1 \\ 1 & \text{if } y > 1 \end{cases}.$$

(iii) Since $F_{Y_n}(y)$ does not actually depend on n , the limiting cdf is simply

$$F_Y(y) = \lim_{n \rightarrow \infty} F_{Y_n}(y) = \begin{cases} 0 & \text{if } y < 0 \\ y & \text{if } 0 \leq y \leq 1 \\ 1 & \text{if } y > 1 \end{cases}.$$

(iv) The limiting cdf $F_Y(y)$ is also the cdf of a $\text{Unif}(0, 1)$ random variable, so $Y_n \xrightarrow{d} Y$ for $Y \sim \text{Unif}(0, 1)$.

Exercise 4 (mle for uniform distribution). Let $X_1, \dots, X_n$ be iid RVRVs from a uniform distribution $\text{Unif}(0, \theta)$ with unknown parameter $\theta > 0$. The pdf of X_i is $p(x_i | \theta) = \frac{1}{\theta} \chi_{[0, \theta]}(x_i)$.

- (i) Write down the likelihood function $L(\theta | x)$ for an observed sample $x = (x_1, \dots, x_n)$. Be careful with the indicator functions.
- (ii) Find the maximum likelihood estimator (mle) $\hat{\theta}_{\text{mle}}$ for θ .
[Hint: The likelihood function may not be differentiable with respect to θ , but the maximum value can be found by reasoning about monotonicity and where it is non-zero.]
- (iii) Calculate the expected value $\mathbb{E}[\hat{\theta}_{\text{mle}}]$. Is $\hat{\theta}_{\text{mle}}$ an unbiased estimator for θ ?
[Hint: First find the cdf of $\hat{\theta}_{\text{mle}}$, then its pdf, and then compute the expectation.]

Solution:

(i) The likelihood function for an observed sample $x = (x_1, \dots, x_n)$ is

$$L(\theta | x) = \prod_{i=1}^n p(x_i | \theta) = \prod_{i=1}^n \frac{1}{\theta} \chi_{[0, \theta]}(x_i) = \left(\frac{1}{\theta}\right)^n \prod_{i=1}^n \chi_{[0, \theta]}(x_i).$$

The product of indicator functions $\prod_{i=1}^n \chi_{[0, \theta]}(x_i)$ is 1 if and only if $0 \leq x_i \leq \theta$ for all $i \in [n]$. This is equivalent to $0 \leq \min\{x_1, \dots, x_n\} =: x_{\min}$ and $x_{\max} := \max\{x_1, \dots, x_n\} \leq \theta$. Since $x_i \geq 0$ for all x_i (they come from $\text{Unif}(0, \theta)$), we thus find $L(\theta | x) = \left(\frac{1}{\theta}\right)^n \chi_{[x_{\max}, \infty)}(\theta)$ when all $x_i \geq 0$ (and zero otherwise), or equivalently

$$L(\theta | x) = \begin{cases} (1/\theta)^n & \text{if } \theta \geq x_{\max} \text{ and all } x_i \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

(We assume $x_{\max} > 0$ to rule out the case $\theta = 0$.)

- (ii) Since $L(\theta | x) = 0$ when $\theta < x_{\max}$ and it is monotonically decreasing in θ for $\theta \geq x_{\max}$ its maximum is at the smallest value θ for which the likelihood is non-zero. This is at $\theta = x_{\max}$. The mle is $\hat{\theta}_{\text{mle}} = X_{\max} = \max\{X_1, \dots, X_n\}$.

In the “bus-waiting-example,” this means that if you’ve been waiting for the bus for (4, 1, 7, 12, 5, 9, 10, 2) minutes on consecutive work days and you model bus arrival time as uniform, then the maximum likelihood estimate of the intervals in which the bus goes (maximum waiting time) is 12 minutes.

- (iii) To find $\mathbb{E}[\hat{\theta}_{\text{mle}}] = \mathbb{E}[X_{\max}]$, we first find the cdf of $X_{\max}$

$$F_{X_{\max}}(t) = P(X_{\max} \leq t) = P(\max\{X_1, \dots, X_n\} \leq t).$$

Since X_i are iid, $P(X_{\max} \leq t) = P(X_1 \leq t, X_2 \leq t, \dots, X_n \leq t) = \prod_{i=1}^n P(X_i \leq t)$. The cdf of $X_i \sim \text{Unif}(0, \theta)$ is $F_{X_i}(x) = x/\theta$ for $0 \leq x \leq \theta$, 0 for $x < 0$, and 1 for $x > \theta$. So, for $0 \leq t \leq \theta$, $F_{X_{\max}}(t) = (t/\theta)^n$. The pdf of $X_{\max}$ is

$$p_{X_{\max}}(t) = \frac{d}{dt} F_{X_{\max}}(t) = \begin{cases} n \frac{t^{n-1}}{\theta^n} & \text{if } 0 \leq t \leq \theta, \\ 0 & \text{otherwise.} \end{cases}$$

Now, we compute the expectation via integration

$$\begin{aligned}
 \mathbb{E}[X_{\max}] &= \int_0^\theta t \cdot p_{X_{\max}}(t) dt \\
 &= \int_0^\theta t \cdot n \frac{t^{n-1}}{\theta^n} dt \\
 &= \frac{n}{\theta^n} \int_0^\theta t^n dt \\
 &= \frac{n}{\theta^n} \left[\frac{t^{n+1}}{n+1} \right]_0^\theta \\
 &= \frac{n}{\theta^n} \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1} \theta.
 \end{aligned}$$

The bias is $\text{Bias}[\hat{\theta}_{\text{mle}}] = \mathbb{E}[\hat{\theta}_{\text{mle}}] - \theta = \frac{n}{n+1}\theta - \theta = -\frac{\theta}{n+1}$. Since the bias is not zero (for finite n), $\hat{\theta}_{\text{mle}} = X_{\max}$ is a biased estimator for θ . It is asymptotically unbiased since $\text{bias}[\hat{\theta}_{\text{mle}}] \rightarrow 0$ as $n \rightarrow \infty$.